

SMALL BUSINESS SALES & OPERATIONS

Data Analysis Report

Portfolio demonstration of data cleaning, analysis, visualization, and business insight generation.

Prepared by: Dennis Maaga

Dataset: Synthetic retail/e-commerce sales data

Period: January–December 2025

Executive Summary

The dataset contains 360 synthetic orders. Total revenue was **\$14,303.85**, generating **\$6,450.85** gross profit at an overall margin of **45.1%**. The analysis examines sales trends, category performance, and sales-channel differences.

1. Data Preparation

Order-level records were structured by date, product, category, channel, region, quantity, discount, revenue, cost, and profit. A calculated margin field was added. In a real engagement, this workflow would also include duplicate checks, missing-value review, data-type validation, and reconciliation with source records.

2. Key Performance Indicators

Metric	Result
Orders	360
Units sold	527
Revenue	\$14,303.85
Gross profit	\$6,450.85
Overall margin	45.1%
Average order value	\$39.73

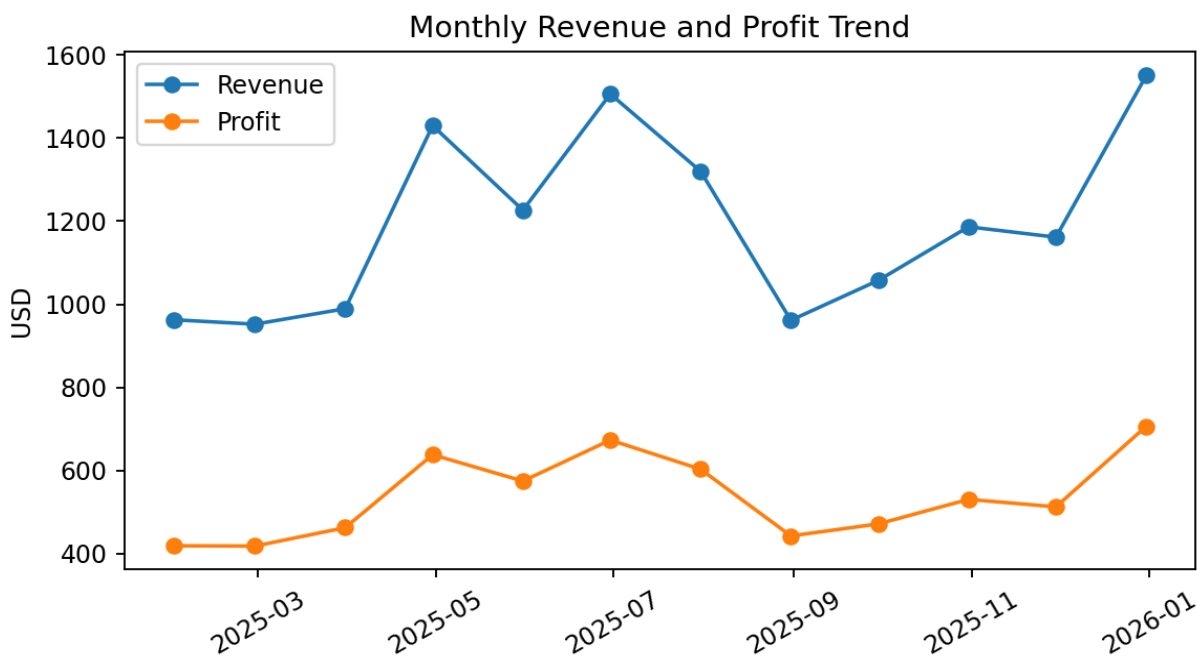


Figure 1. Monthly revenue and profit trend.

3. Product Category Performance

Category	Revenue	Profit	Margin
Electronics	\$6,694	\$2,806	41.9%
Home Office	\$5,148	\$2,285	44.4%
Stationery	\$2,462	\$1,360	55.2%

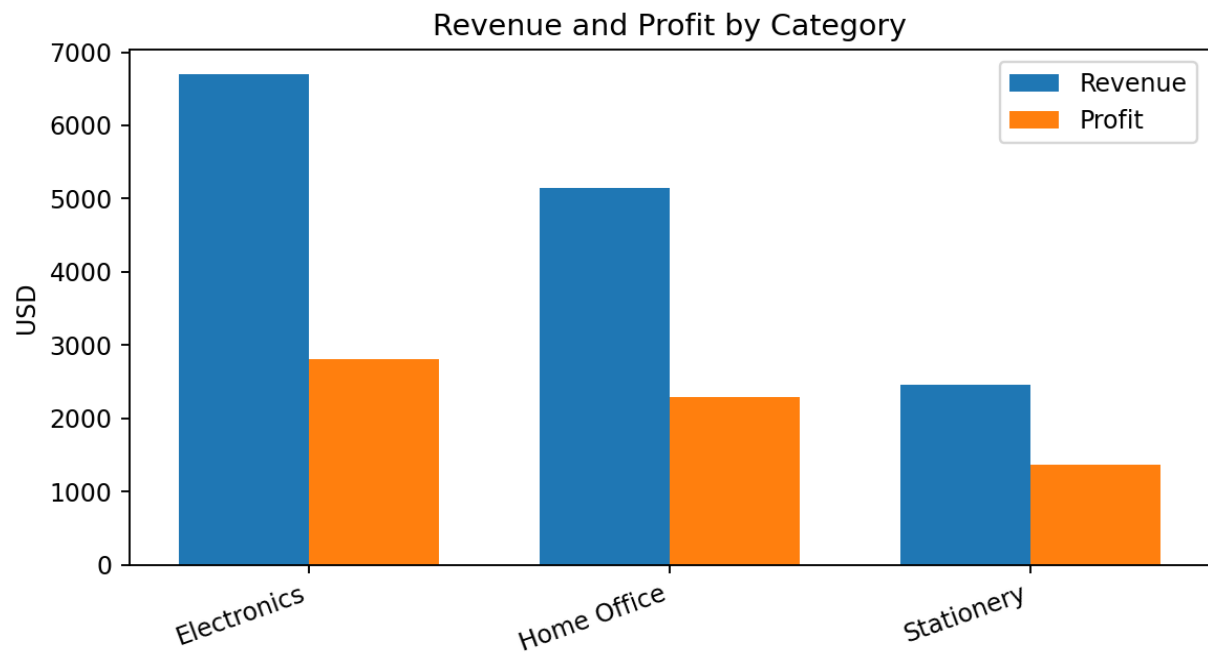


Figure 2. Revenue and profit by product category.

4. Sales Channel Analysis

Channel	Revenue	Profit	Margin
Website	\$6,109	\$2,756	45.1%
Marketplace	\$5,991	\$2,681	44.8%
Wholesale	\$2,203	\$1,013	46.0%

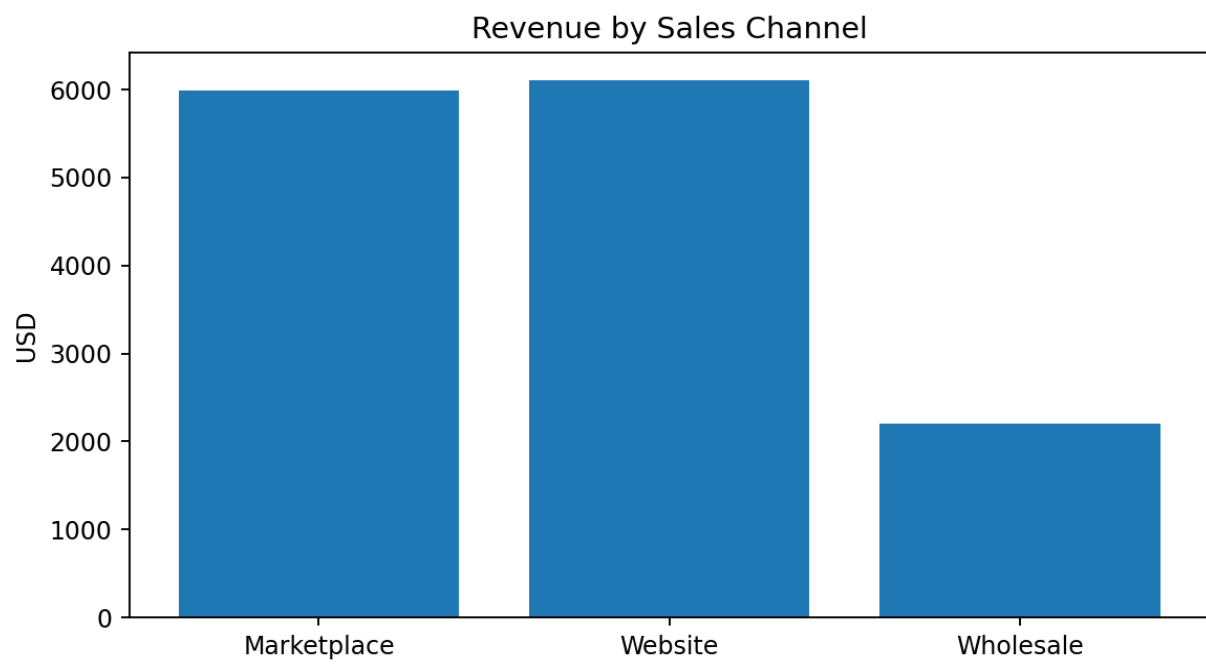


Figure 3. Revenue contribution by sales channel.

5. Business Insights

Category concentration: Electronics generated the highest revenue in the illustrative dataset, making it important for inventory and promotion decisions.

Channel contribution: Website produced the highest total revenue, while Wholesale recorded the strongest gross margin among the channels.

Seasonality: Q4 revenue was \$3,898, compared with \$2,904 in Q1, showing a stronger final-quarter period in the synthetic dataset.

Profitability: Revenue leadership and margin leadership are not necessarily the same; decisions should consider both sales volume and cost structure.

6. Recommendations

- Prioritize stock availability for consistently high-revenue categories.
- Review discounts by product and channel to identify promotions that grow volume without unnecessarily reducing margin.
- Track revenue, gross profit, margin, average order value, and units sold in a recurring dashboard.
- Investigate underperforming products before discontinuing them; low sales can reflect pricing, visibility, or channel placement.
- Compare monthly performance against targets and prior periods.

7. Conclusion

This sample demonstrates how raw order-level information can be transformed into decision-ready business insight. The workflow is applicable to small retailers, e-commerce businesses, distributors, and other organizations that need practical analysis from spreadsheet data.

Portfolio note: all transactions and business details are synthetic and created solely for demonstration.